SYLLABUS

Chapter 2 (Kothari & Nagrath) — Series Impedance of Transmission Lines

1. Resistance
2. Inductance
3. Inductance of a Conductor due to Internal Flux
4. Flux Linkage Between two points external to an isolated conductor
5. Inductance of a single-phase two-wire line
6. Flux linkages of one conductor in a group
7. Inductance of Composite-Conductor lines
8. Inductance of three-phase lines with equilateral spacing
9. Inductance of three-phase lines with unsymmetrical spacing
10. Bundled Conductors

Chapter 3 (Kothari & Nagrath) — Capacitance of Transmission Lines

1. Electric field of a long straight conductor
2. The potential difference between two points due to a charge
3. Capacitance of a two-wire line
4. Capacitance of a three-phase line with equilateral spacing
5. Capacitance of a three-phase line with unsymmetrical spacing
6. Effect of earth on the capacitance of three-phase transmission line
7. Bundled Conductors

Chapter 11 (Mehta) — Underground Cables

1. Construction of cables
2. Insulating material of cables
3. Classification of cables
4. Cables for three-phase service
5. Laying of cables
6. Insulating resistance of a single-core cable
7. Capacitance of a single core cable
8. Dielectric stress of single core cable
9. Grading of cable
10. Capacitance grading
11. Inter-sheath grading
12. Capacitance of 3-core cable
13. Measurement of Ce and Cc

14. Current carrying capacity of the cable
15. Thermal resistance
16. Permissible current loading
17. Types of the cable fault

Chapter 8 (Mehta) — Mechanical Design of Overhead Line

1. Main components of overhead line
2. Conductor Material
3. Line Support
4. Insulators
5. Types of Insulator
6. Potential Distribution over Suspension insulator String
7. String efficiency
8. Method of improving string efficiency
9. Corona
10. Sag in overhead Line
11. Calculation of Sag

Chapter 12 (Kothari & Nagrath) — Stability

1. Swing equation
2. Synchronous machine models for stability studies
3. Steady –State Stability
4. Transient- Stability- Equal area criterion
5. Application to a sudden increase in power input
6. Application to a three-phase fault
7. Numerical solution of Swing equation
8. Multi-machine system
9. Multi-machine transient stability
10. Factors affecting stability

Compensation in Power System

1. load compensation, line compensation, series compensation, shunt compensators, flexible ac transmission systems (facts), and facts controllers.

Flexible AC Transmission Systems (FACTs)

1. FACTs devices and Applications.

High Voltage DC Transmission System

1. Comparison of DC and AC system, Application of DC Transmission System, Types of HVDC system, HVDC system configuration and components, Control of HVDC system, Response to DC and AC system faults.

Power Quality

1. Electromagnetic phenomena and power quality; temporary phenomena (transient, long duration voltage variations, sustained interruptions, short duration voltage variations), Interruption, Voltage sags and swells, harmonics. Steady-state phenomena; voltage imbalance, waveform distortion, voltage fluctuation and flicker, and power frequency variations.